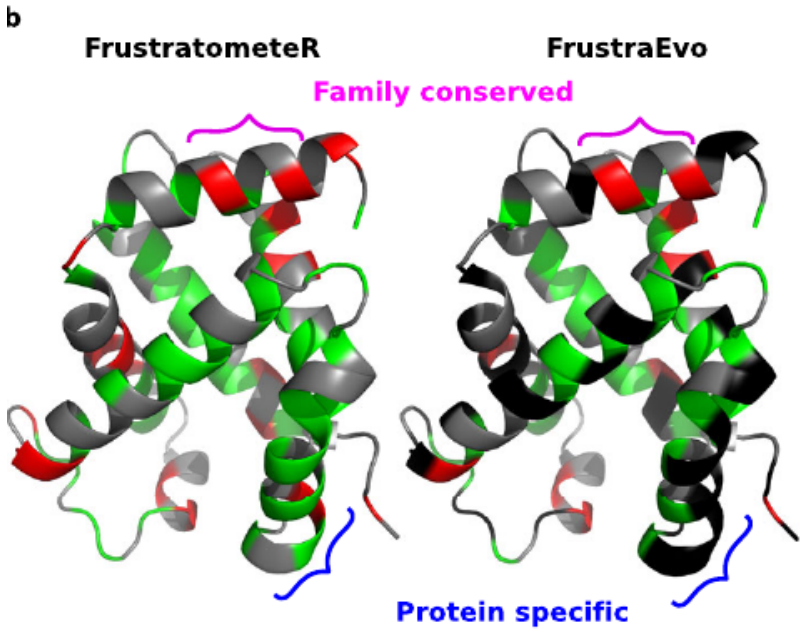
Local energetic frustration conservation in protein families and superfamilies

Maria I. Freiberger, Victoria Ruiz-Serra, Camila Pontes, Miguel Romero-Durana, Pablo Galaz-Davison, Cesar A. Ramírez-Sarmiento, Claudio D. Schuster, Marcelo A. Marti, Peter G. Wolynes, Diego U. Ferreiro, R. Gonzalo Parra and Alfonso Valencia

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Affiliations: Universidad de Buenos Aires (IQUIBICEN/CONICET), Barcelona Supercomputing Center, Pontificia Universidad Católica de Chile, Rice University, ICREA



Presenter: Buse Giledereli

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Introduction and Motivation

Executive summary

- **Primary focus:** FrustraEvo, an automated framework that quantifies how conserved local energetic frustration is across protein families and superfamilies.
- **Core idea:** mapping frustration onto multiple sequence alignments (MSAs) connects sequence conservation with energy landscape theory.
- This method showcases constraints needed for **folding and stability** and energetic conflicts conserved for **function, allostery and binding specificity**.

Stability and binding prediction

FrustIC vs ddPCA stability and binding for SH3, PDZ and KRAS

Functional divergence

α vs β globins; the human RAS superfamily

Proteome-wide annotation

SARS-CoV-2 across the coronavirus phylogeny

Protein engineering

Triggering the fold switch of metamorphic RfaH

Theoretical Background

Local frustration: what it is and how it is calculated

- **Minimal frustration:** native proteins minimize energetic conflicts, giving a funnel-shaped landscape that guides folding.
- **Functional frustration:** some conflicts (frustration) are kept by selection at active sites, binding interfaces and allosteric sites.
- **Evolutionary view:** sequence diverges faster than structure, so frustration can be compared across a whole family.

~40%	**Minimal:** stabilizing; builds the core
~50%	**Neutral:** neither favorable nor conflicting
~10%	**High:** penalty enabling flexibility, binding or reactivity

How local frustration is calculated (FrustratometeR)

1. **Define contacts** by Cβ distance: short-range (< 6.5 Å), long-range (6.5–9.5 Å) or water-mediated (long-range and solvent-exposed). The type selects the AWSEM energy term.
2. **Score the native energy** of each contact or residue with the coarse-grained AWSEM potential. Only the 20 amino acids are parameterized, so **ligands are invisible**.
3. **Generate 2,000 decoys.** Mutational: swap in random residues for the pair in place. Configurational: also vary distance and burial. Single residue: swap one residue, summing over all its contacts.
4. **Normalize** as a Z-score (≈ –4 to 4), then classify
   FI < –1 highly frustrated; > 0.78 (pairs) or > 0.55 (single residue) minimally frustrated.

Because substrates and partner proteins are not in the energy function, binding sites can appear frustrated in isolated monomers.

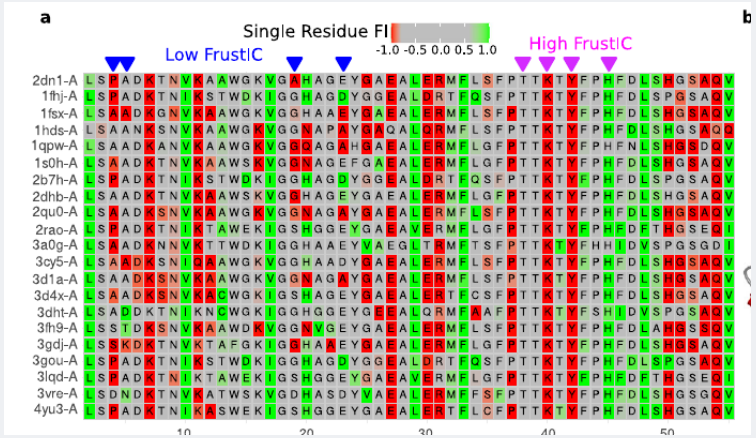
Sources: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Methods; Ferreiro et al., *PNAS* 104, 19819 (2007)

Modes of Operation

Single residue vs. contacts mode

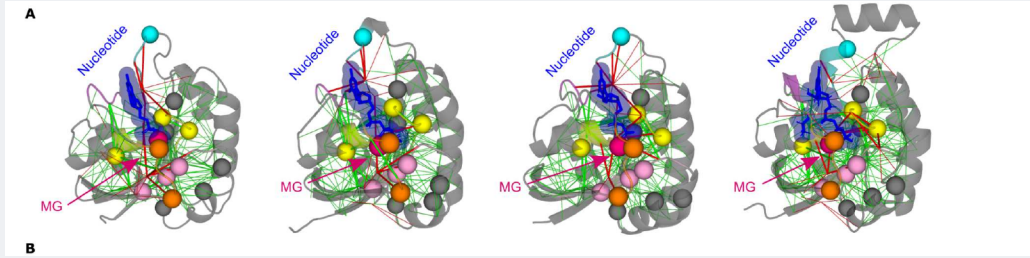
Single residue mode (SRFI)

- Integrates every interaction a residue makes with its environment.
- Outputs SeqIC/FrustIC logos, Multiple Sequence Frustration Alignments (MSFA) and consensus MSFAs.



Contacts mode (mutational / configurational)

- **Mutational FI:** identities of the pair are varied, positions fixed: sequence–structure fit.
- **Configurational FI:** identities, distance and solvent exposure varied: packing and environment.
- Best for divergent superfamilies where identities drift but contact geometry is kept.



Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Fig. 1a (left) and Supplementary Fig. 5A (right)

Methodological Framework

The FrustraEvo tool

Pipeline overview

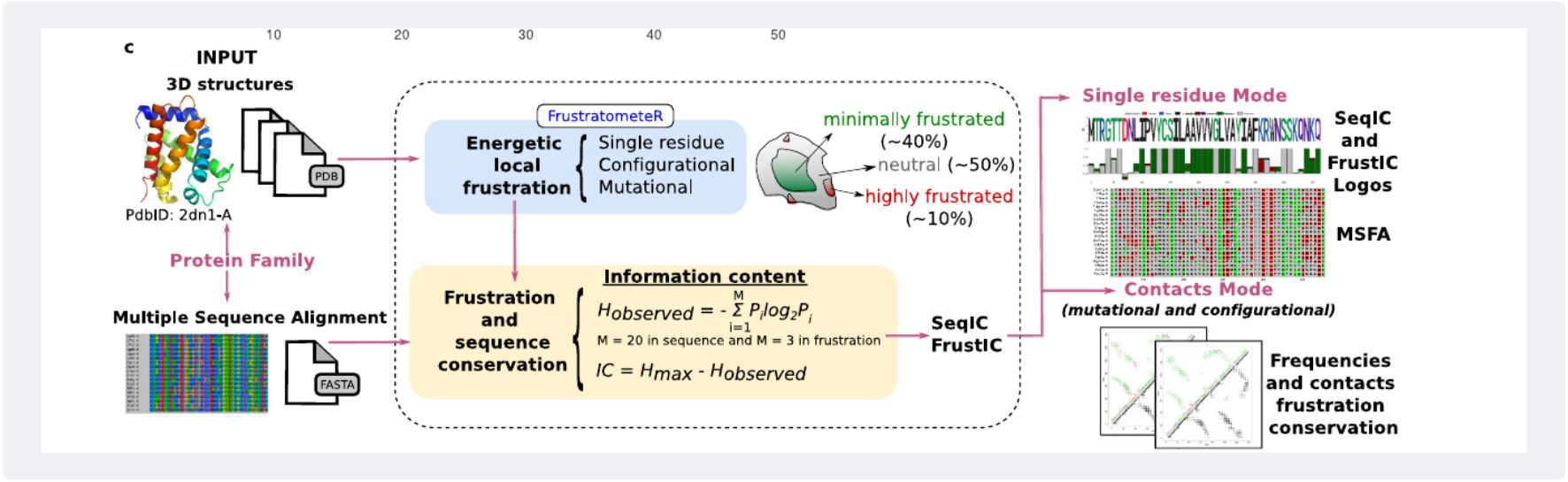
Input: an MSA (FASTA) plus one 3D structure per sequence: experimental PDB or AlphaFold2 model.

Engine: local frustration from FrustratometeR, using the coarse-grained AWSEM energy function.

Decoys and Frustration Index

$$FI = (\langle Edecoy \rangle - Enative) / \sigma decoy$$

Positive FI: native more favorable (less energy) than decoys (minimally frustrated). **Negative FI:** native higher energy than decoys (highly frustrated).



Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Fig. 1c

Quantification of Conservation

From per-protein frustration (SRFI) to family-level conservation (FrustIC)

① SRFI / pairwise FI: is a residue frustrated?

ONE STRUCTURE · ONE RESIDUE OR PAIR

Z-score of the native energy against 2,000 decoys. **Continuous**, in SD units (≈ –4 to 4), then binned into a state:

State	SRFI	Pairwise FI
Highly frustrated	< –1.0	< –1.0
Neutral	–1.0 to 0.55	–1.0 to 0.78
Minimally frustrated	> 0.55	> 0.78

> **Answers:** how frustrated is this residue *in this protein*?

② FrustIC: is that frustration conserved?

WHOLE FAMILY · ONE MSA COLUMN

Takes the **states** from ① for every structure at an aligned column and measures how uniform they are. In **bits**.

$$Hobs = -\Sigma\, Pi \log_2 Pi \;(M = 3 \text{ states})$$

$$\text{FrustIC} = Hmax - Hobs$$

Background $Pmin$ 0.4, $Pneu$ 0.5, $Pmax$ 0.1; ceiling ≈ $\log_2 3 = 1.58$; **conserved if FrustIC > 0.5**.

> **Answers:** is the frustration state *the same across the family*?

- high FrustIC ≠ high frustration.
- SeqIC applies the same formula to amino acid identities (M = 20).

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Methods

Results I: Benchmark and Validation

Setup: can FrustIC predict measured stability and binding?

QUESTION

Does conserved frustration correlate with stability and binding?

Experimental reference

- **ddPCA** (double-deep protein fragment complementation) scores the effect of each single mutant on **abundance** (stability) and **binding** (function).

Protein	Family set	Mean pLDDT
GRB2-SH3	173 homologs	89.3
PSD95-PDZ3	679 homologs	78.3
KRAS	36 curated (Rojas et al.) + automatic set	81.6 (auto)

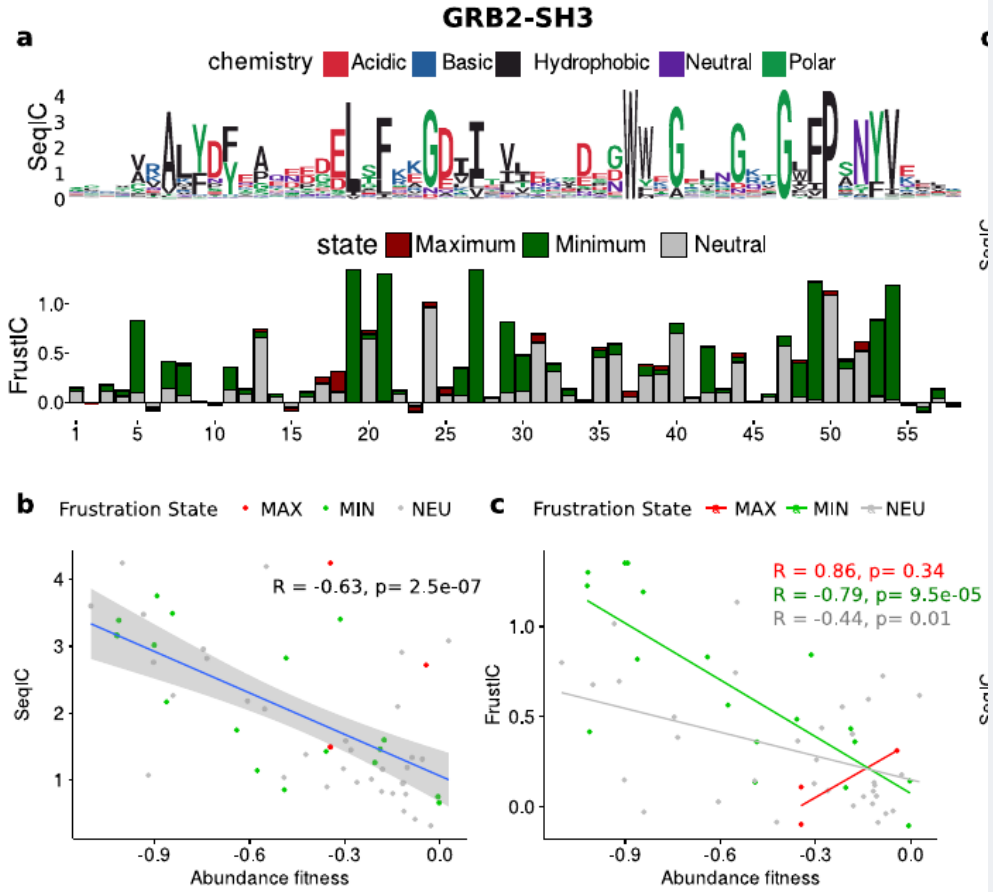
Approach

1. Retrieve homologs (BLAST, quality filters, CD-HIT), align and model each with AlphaFold2.
2. Run FrustraEvo (single residue mode): SeqIC and FrustIC for every MSA column.
3. Average the ddPCA fitness of all mutants at each position.
4. Correlate SeqIC and FrustIC with fitness (Pearson), split by conserved frustration state.

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Results and Methods

Results I: Stability GRB2-SH3

Conserved minimal frustration tracks stability

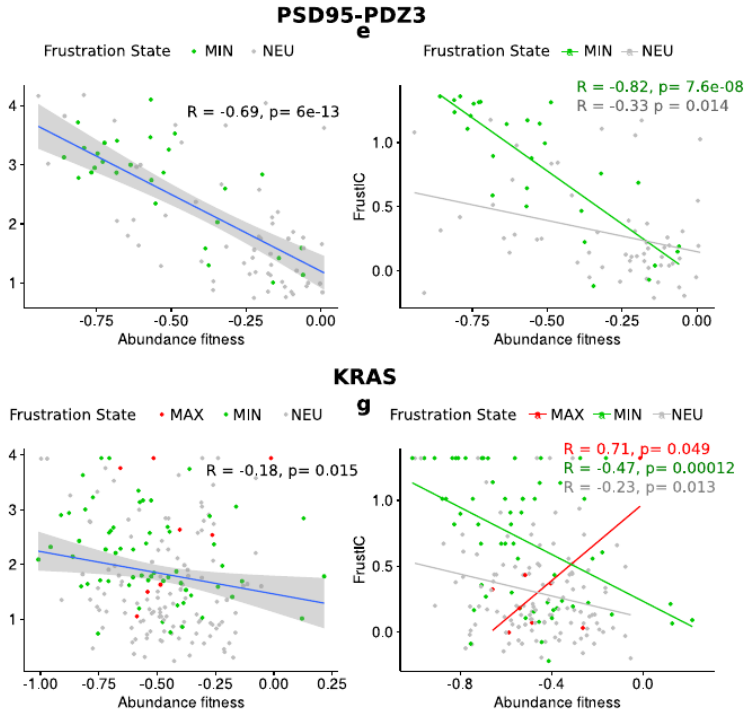


- **Logo plot (a):** most frustrationally conserved columns are minimally frustrated (green); SH3 has **no** conserved highly frustrated position.
- **Why:** SH3 binds many different partners, so frustrated interface residues are not aligned across the family and their signal averages out.
- **Stability (b vs c):** SeqIC r = –0.63, while FrustIC of minimally frustrated positions reaches r = –0.79 with abundance fitness.
- **Why FrustIC does better:** core positions swap between hydrophobic residues (e.g., Leu↔Val) without losing minimal frustration, so their frustration state stays conserved while their sequence drifts.

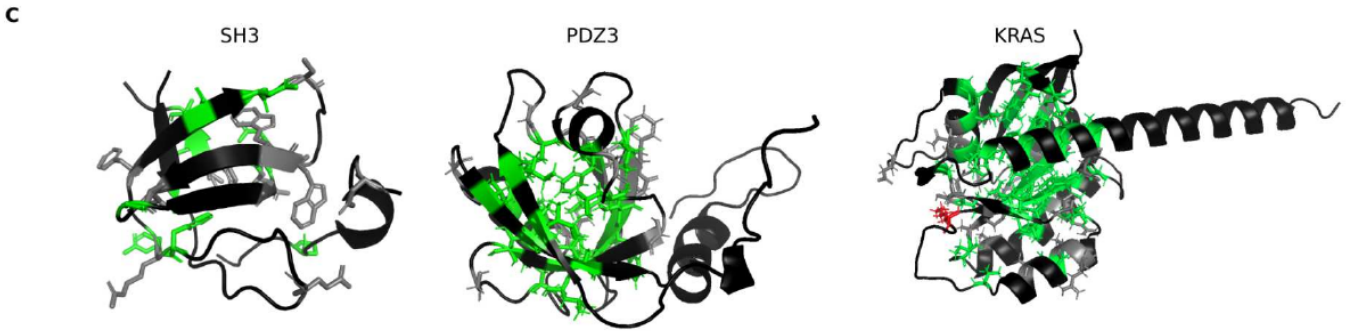
Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Fig. 2a–c

Results I: Stability PSD95-PDZ3 and KRAS

A localized function weakens the correlation



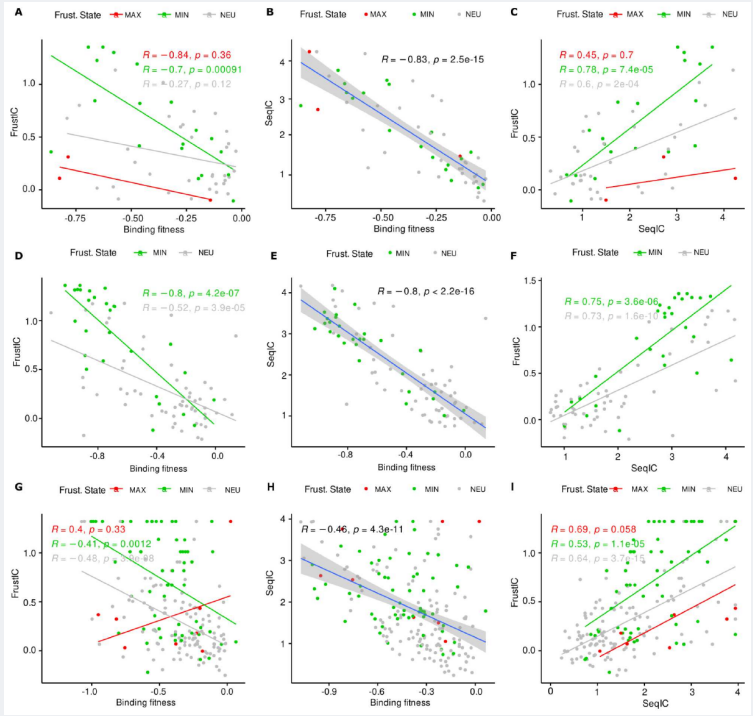
- **PDZ3 (d, e; 679 homologs):** FrustIC r = –0.82 vs SeqIC r = –0.69. As in SH3, no conserved highly frustrated positions.
- **KRAS (f, g; curated 36-protein set):** SeqIC barely relates to stability (r = –0.18); FrustIC of minimally frustrated positions gives r = –0.47.
- **K117** is the only conserved, highly frustrated position. It is one of seven nucleotide-contacting residues, and most mutations there *increase* abundance: function conflicts with stability.
- The nucleotide is invisible to the energy function, among all pocket residues the analysis singles out K117, matching prior experiments showing K117's functional importance.
- Positions with FrustIC near the ceiling (~1.58) sit in the hydrophobic core (colored where FrustIC > 0.5).

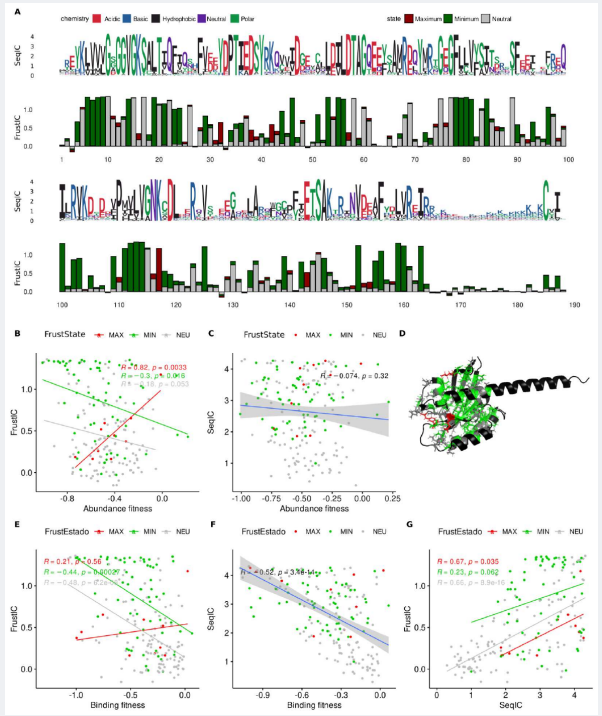


Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Fig. 2d–g and Supplementary Fig. 1C

Results I: Binding Fitness

Supplementary validation





- **Binding (Supp. Fig. 2):** FrustIC and SeqIC also relate to binding fitness, and to each other, in all three systems.
- **Automatic KRAS set (Supp. Fig. 3):** 1,354 BLAST-retrieved homologs reproduce the trends, but the FrustIC–abundance correlation is weaker.
- Phylogeny-defined families beat simply filtered BLAST hits: **curation matters**.
- r ≈ 0.8 (SH3, PDZ3) and 0.47 (KRAS): not a full stability model, but a useful in silico proxy.

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Supplementary Figs. 2–3

Results II: Functional Divergence in Related Families

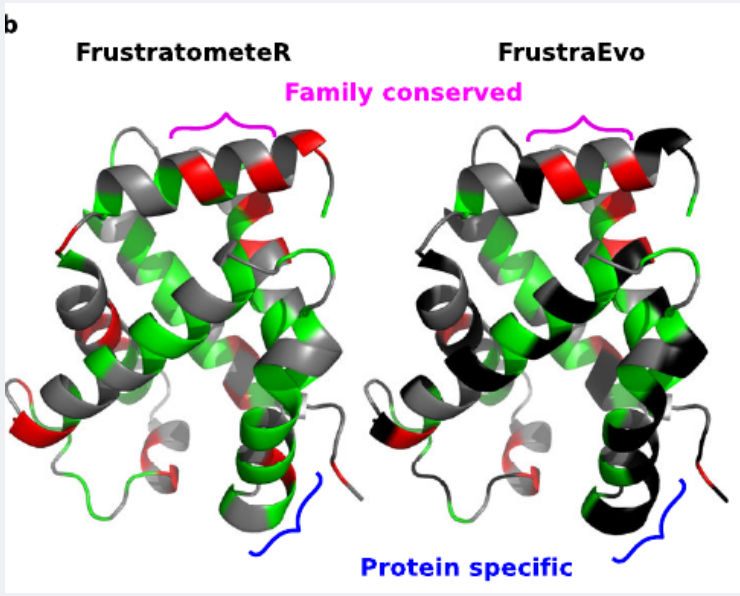
Setup: α vs. β globins

> QUESTION
>
> Same fold, different jobs in hemoglobin: does frustration diverge where function diverges?

- **Data:** 21 non-redundant mammalian hemoglobin crystal structures (PDB, April 2022), split into α and β chains; reference human hemoglobin PDB 2DN1.

Approach

1. FrustratometeR on every chain; three MSAs: α+β, α only, β only.
2. FrustraEvo in single residue mode → SeqIC and FrustIC logos.
3. S3Det SDPs between α and β; compare their frustration states.
4. Map family-specific, highly frustrated positions to known functional sites.

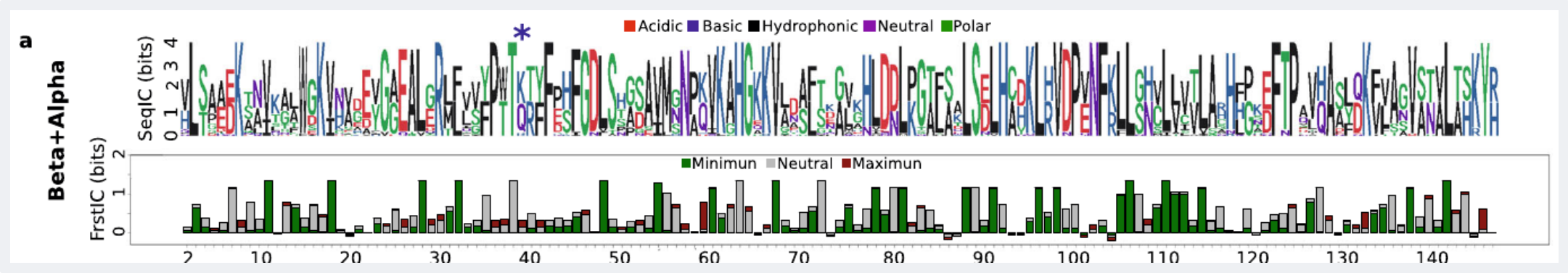


Human α globin: frustration in one structure (left) vs frustration conserved across the α family (right; black = not conserved).

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Fig. 1b and Methods

Results II: Combined α/β Globin Alignment

Shared constraints and specificity determining positions



- Conserved frustration is dominated by **minimally frustrated** (n = 35, mean FrustIC 1.02) and **neutral** positions (n = 34, mean 0.85); only 3 are highly frustrated (mean 0.72).
- S3Det finds **15 SDPs (Specificity Determining Position)** between α and β; **6** differ in frustration state..

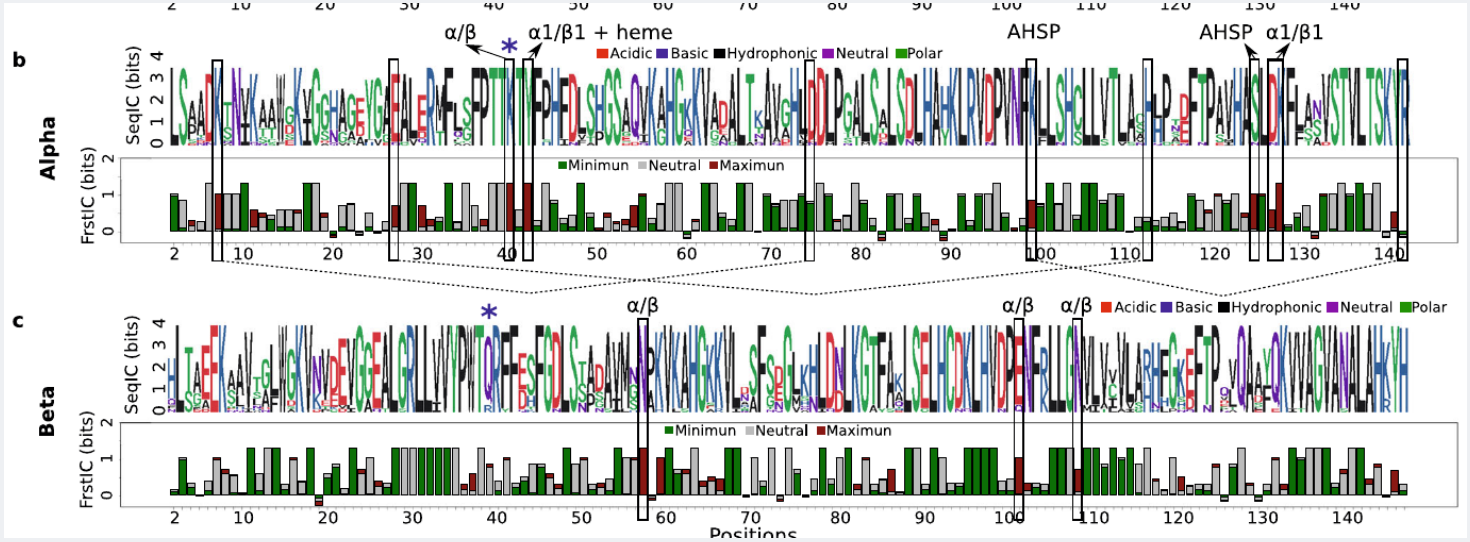
MSA position	α globin	β globin
32	Phe, minimal	Leu, minimal
57	Ser, neutral	Asn, highly
140	Val, minimal	Ala, neutral

Identity can change while energy is kept (32), or energy can diverge (57, 140).

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Fig. 3a

Results II: Family-Specific Frustration

Separate α and β alignments



b: α only; c: β only. Rectangles mark functionally relevant positions; asterisk marks MSA position 39.

- **12** conserved highly frustrated positions in α (mean FrustIC 0.87) and **8** in β (0.88); only **2** are shared (Q54α/K59β, Y140α/Y145β).
- Several sit at the asymmetric subunit contacts of the tetramer: K40α, Y42α; W37β, N57β, E101β, N108β.

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Fig. 3b–c

Results II: Known functional residues and FrustIC

Known functional residues are frustrationally conserved

Residue	FrustIC	Known role
Lys7α	1.02	Intra-α salt bridge with Asp74α
Lys40α	1.29	α1β2 salt bridge with His146β; cooperativity
Tyr42α	1.29	α1β2 H-bond with Asp99β; T-state, heme contact
Lys99α	0.84	AHSP binding; disruptive mutations cause anemia
Ser124α	1.02	AHSP binding; Hb Policoro (Pro) → α-thalassemia
Lys127α	1.29	Hb Waikato (Gln) raises O_2 affinity
Trp37β	0.59	α1β2 contacts; mutations shift quaternary state
Asn57β	1.29	Hb G-Ferrara: higher O_2 affinity at acid pH
Glu101β / Asn108β	1.02 / 0.70	Interface with α globin
Tyr145β	0.68	Possible electron transfer to heme

Several residues (K40α, Y42α, W37β, E101β) sit on the α/β interface that switches between T and R states (Bohr effect).

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Supplementary Table (functional residues in hemoglobins)

Results III: Superfamily Diversification

Setup: the human RAS superfamily

QUESTION

When sequences have diverged widely, which energetic constraints survive across the superfamily, and which are subfamily-specific?

- Small GTPases share one fold and GTP hydrolysis; the GTP site is built from five motifs, **G1–G5**. Subfamilies bind different effectors.
- 160 human paralogs from Rojas et al. (2012); 9 no longer in UniProt; RAN (1 sequence) excluded.

ARF

26 sequences · mean pLDDT 86.3

RAB

64 sequences · mean pLDDT 77.4

RAS

38 sequences · mean pLDDT 83.5

RHO

22 sequences · mean pLDDT 82.6

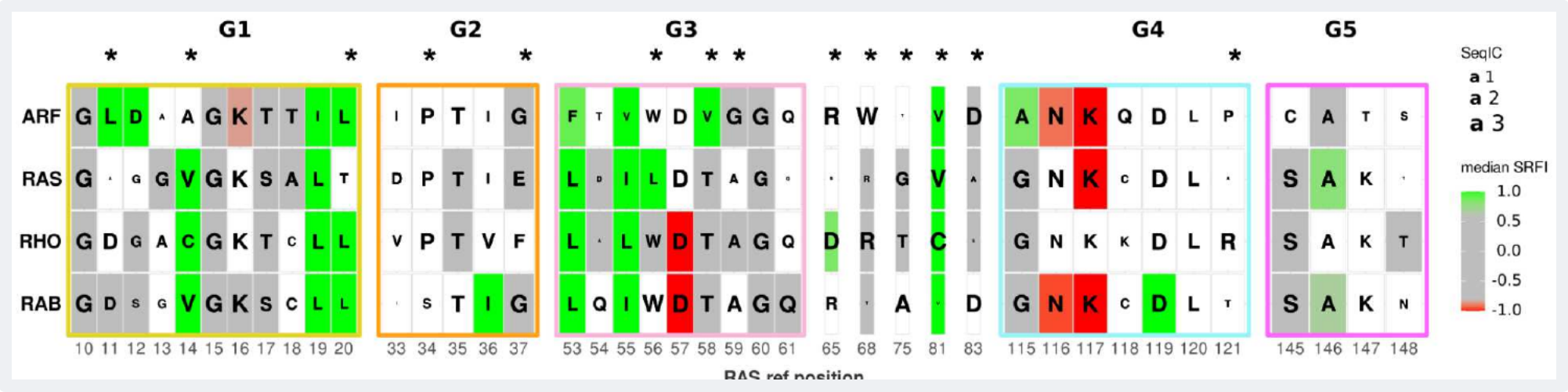
Approach

1. Align with MAFFT; model every sequence with AlphaFold2.
2. Single residue mode across the G1–G5 motifs.
3. Contacts mode with the **mutational** FI, since function should depend on amino acid identity.
4. Draw networks of conserved contacts among G1–G5 residues and SDPs.

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Methods and Supplementary Note 1

Results III: RAS Superfamily, Single-Residue View

Consensus MSFA of the G1–G5 motifs



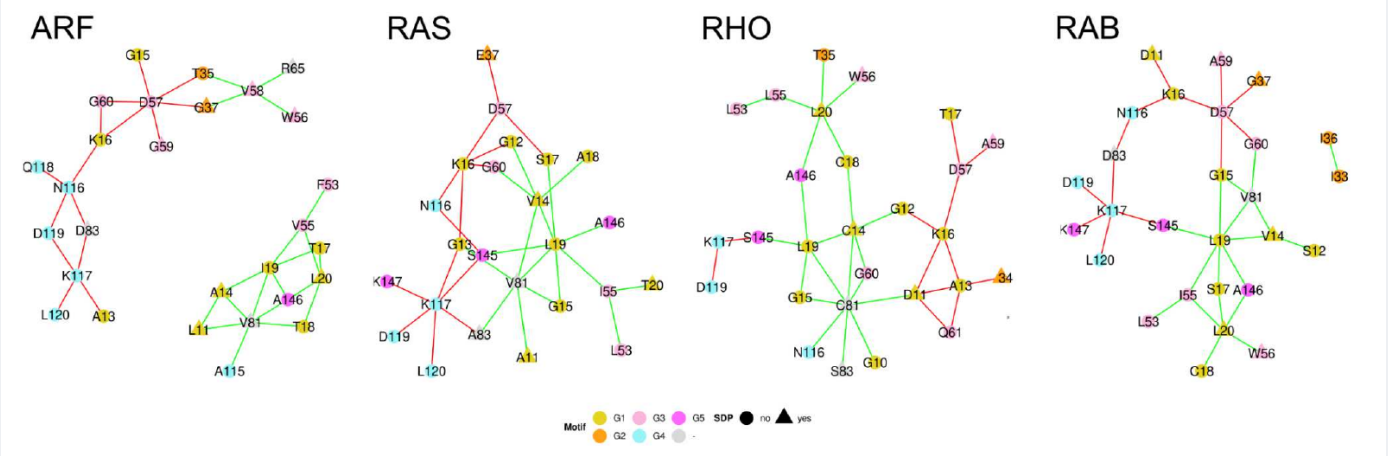
- Minimally frustrated, conserved positions 14, 19, 20, 53, 56 and 81 cluster in G1 and G3; G2, G4 and G5 are less energetically conserved.
- G4 N116 and K117 contact the purine (specificity) and are highly frustrated and conserved.
- T35 (G2; senses GTP, binds Mg^{2+}) is conserved in sequence but not in energy: its loop adopts different conformations.
- So the authors moved to **contacts** with the mutational FI.

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Supplementary Fig. 4 and Supplementary Note 1

Results III: Conserved Contact Networks

Mutational FI across subfamilies

- Each graph shows contacts conserved (FrustIC > 0.5) in ≥ 50% of a subfamily's structures, among G1–G5 residues (circles) and SDPs (triangles).
- A highly frustrated network around the GTP/Mg^{2+} pocket, built mainly from **Lys16 (G1)**, **Asp57 (G3)** and **Lys117 (G4)**, is present in all four subfamilies.
- The substrate is not in the energy function, so its pocket looks frustrated; in an uncharacterized family such a network would flag function.



Specificity Determining Position 83 (RAS numbering)

Subfamily	Residue at 83	Contact frustration
ARF, RAB	Asp (conserved)	Highly frustrated
RHO	Ser (conserved)	Minimally frustrated
RAS	Variable	Mixed

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Supplementary Fig. 5B and Supplementary Note 1

Results IV: Large-Scale Application

Setup: proteome-wide coronavirus analysis

QUESTION

Can the method run unsupervised on a whole proteome and flag constraints in poorly characterized proteins, such as those of emerging pathogens?

1

Homologs and MSAs

BLAST of each SARS-CoV-2 protein against NCBI nr, Coronaviridae only; quality filters; MAFFT; CD-HIT.

2

Subfamilies

S3Det splits each family into subfamilies and finds their SDPs.

3

Structures

AlphaFold2 models trimmed to a SARS-CoV-2 reference PDB; clusters kept if mean pLDDT ≥ 80 and ≥ 10 sequences.

4

Frustration

FrustraEvo in single residue mode for every subfamily.

29

SARS-CoV-2 proteins or domains processed

22

families passing the quality filters

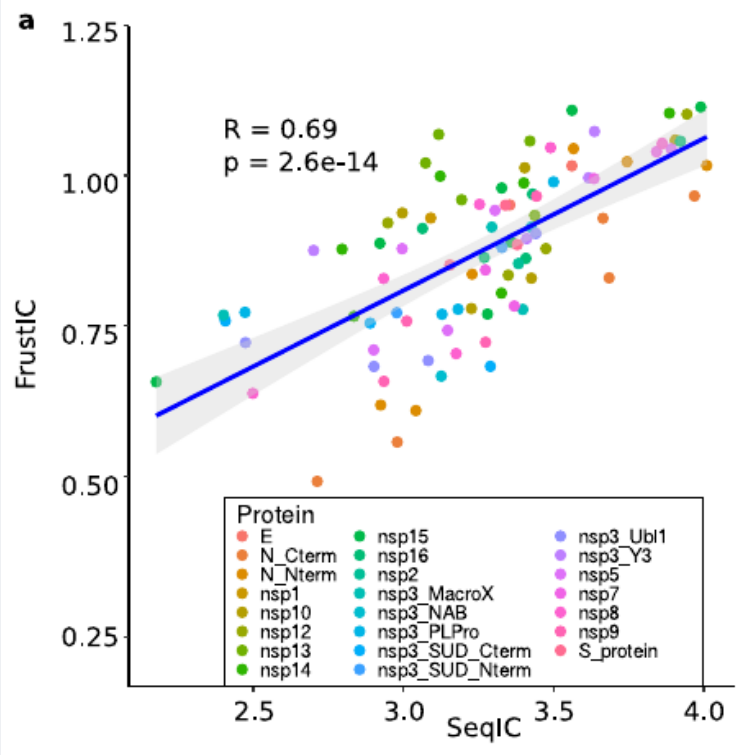
M, nsp4, nsp6

removed: transmembrane, no reference PDB

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Methods and Supplementary Note 2

Results IV: Sequence vs Frustration Conservation

One point per S3Det subfamily

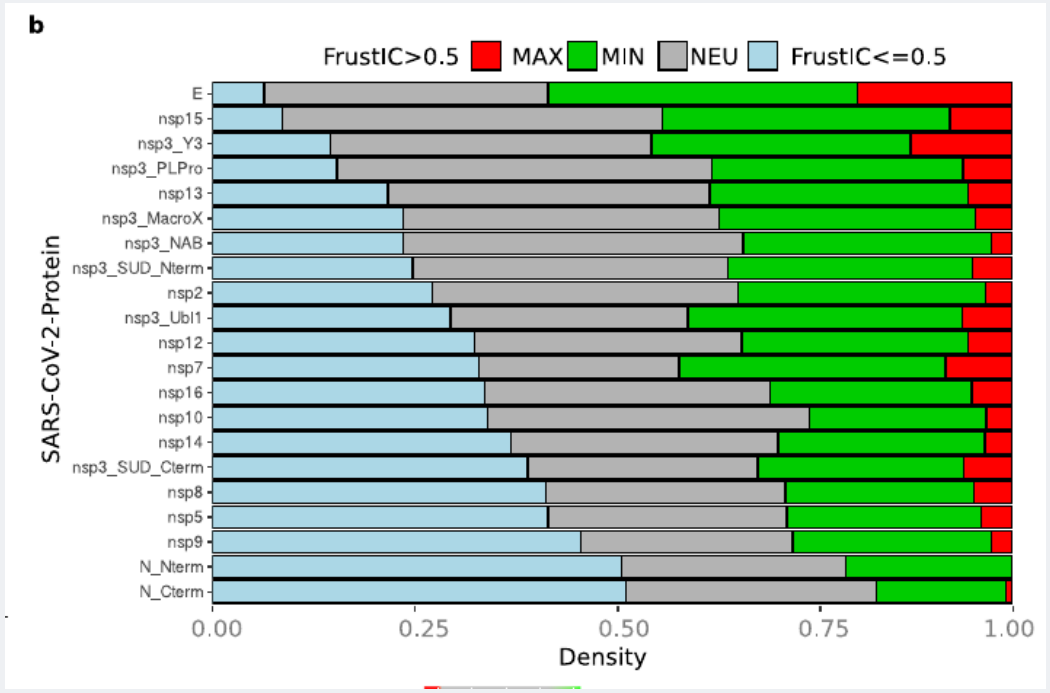


- Mean FrustIC and mean SeqIC per subfamily correlate: **r = 0.69**, p = 2.6×10^{-14}.
- Broadly, families that conserve sequence also conserve frustration states.
- But the spread is large: subfamilies with similar SeqIC can differ widely in FrustIC, so frustration adds information sequence alone does not give.

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Fig. 4a

Results IV: Constraints Across the SARS-CoV-2 Proteome

Sarbecovirus subfamily vs SARS-CoV-2



93.6–48.3%
residues conserved: E protein to N_CTerm

301 / 332
highly frustrated in family, also in SARS-CoV-2

- Among conserved residues, most are neutral (mean proportion 0.35) or minimally frustrated (0.29).
- Of the 332 highly frustrated positions conserved across Sarbecoviruses, 301 (~91%) are also highly frustrated in SARS-CoV-2.
- **62** residues are neutral or minimal in the family but highly frustrated in SARS-CoV-2: candidate recent gains of function (Spike 18, nsp2 9, nsp3 domains 9, nsp5 6).

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Fig. 4b and Supplementary Note 2

Results IV: Case Study, Papain-like Protease (PLPro)

Setup: sites and subfamilies compared

> **QUESTION**
>
> Do PLPro subfamilies differ energetically at sites linked to catalysis and to host immune evasion?

- PLPro cleaves the viral polyproteins and strips ubiquitin (Ub) and ISG15 from host proteins to reduce immune system response.
- 122 homologs split by S3Det into 4 subfamilies matching subgenera: **Sarbeco** (31), **Nobeco** (11), **Merbeco** (35), **Embeco** (45).
- Plus 29 experimental SARS-CoV-2 PLPro structures as a fifth group, capturing one protein's conformational diversity.

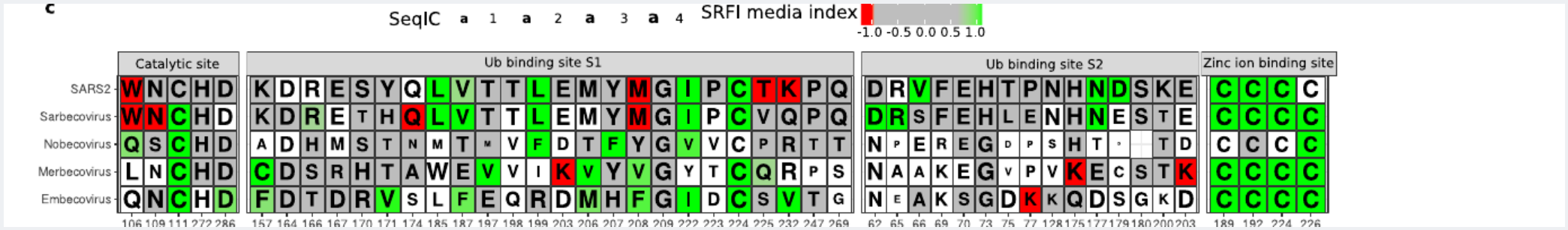
Regions compared

Region	Residues / role
Catalytic site	Trp106 and triad Cys111, His272, Asp286
S1 site	Host-protein binding (ISG15)
S2 site	Host-protein binding (Ub)
Zinc site	Cys189, Cys192, Cys224, Cys226

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Fig. 4c and Methods

Results IV: PLPro Catalytic and Structural Sites

Conserved stability vs subfamily-specific frustration

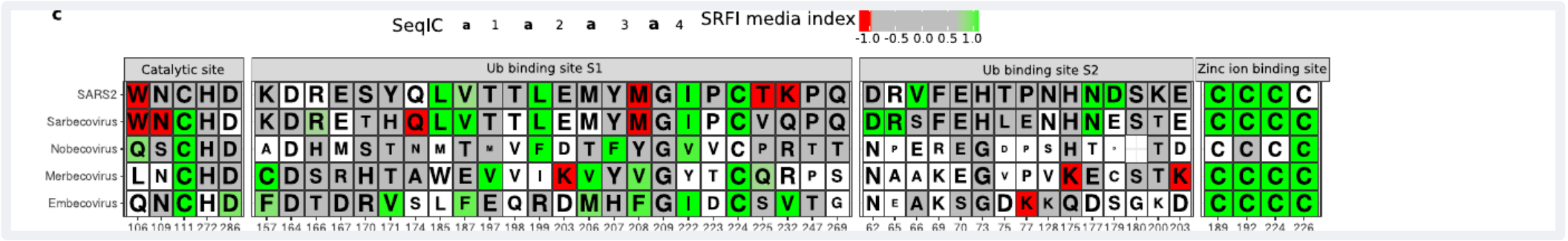


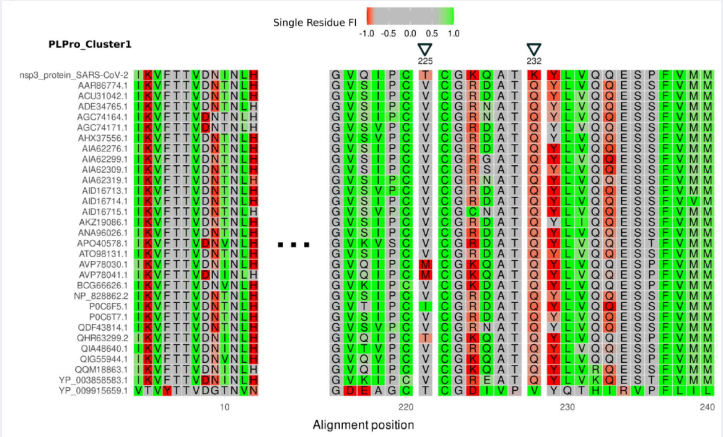
- **Trp106** stabilizes the catalytic triad and is conserved and highly frustrated **only in Sarbecoviruses**.
- Merbecoviruses (MERS) have a Leu106 that is not energetically conserved, and slower catalysis; swapping L→W in MERS enhances catalysis.
- Catalytic **Cys111** is minimally frustrated and conserved in all subfamilies (neutral in the SARS-CoV-2 PDB set because some carry C111S).
- The four **zinc-binding cysteines** are minimally frustrated and conserved in most subfamilies: a stability requirement.

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Fig. 4c

Results IV: PLPro Binding Sites and Emerging Variants

Consensus MSFA and Sarbecovirus detail





- In SARS-CoV-2, the S1 (ISG15) site is more highly frustrated and the S2 (Ub) site more minimally frustrated than in the other viruses.
- Positions **225 / 232**: neutral Val / Gln in Sarbecoviruses, but highly frustrated **Thr / Lys** in SARS-CoV-2; these changes alter Ub association.
- **Thr225** is found only in SARS-CoV-2 and bat RaTG13, its likely progenitor; bat strains Rc-o319 and bat-SL-CoVZXC21 carry a Met that is even more frustrated: a position to watch.
- **Lys232** is unique to SARS-CoV-2 among Sarbecoviruses: a recent gain of function.

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Fig. 4c and Supplementary Fig. 11

Results V: Metamorphic Protein Engineering

Setup: the case of RfaH

QUESTION

What family constraints hold RfaH's C-terminal domain in its α state, and can perturbing them trigger the switch to the β fold?

- RfaH's C-terminal domain (CTD) is an α-hairpin, autoinhibited by binding its N-terminal domain (NTD).
- On RNA polymerase and *ops* DNA, the CTD refolds into a β-barrel.
- Its paralog NusG exists only in the β fold, so metamorphosis likely arose after the two diverged.

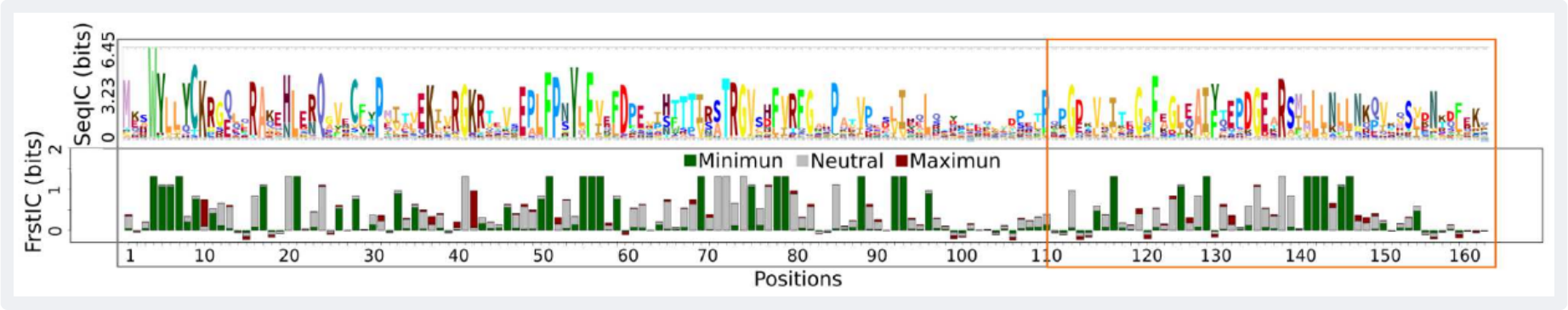
Approach

1. Collect 30 putative metamorphic RfaH orthologs; model them with AlphaFold2.
2. FrustraEvo in single residue and contacts modes (mutational, configurational).
3. Select interface residues between NTD and αCTD.
4. FrustratometeR: frustration change for all 19 substitutions at each residue.
5. Predict designed mutants and NusG-like variants with AlphaFold2.

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Results and Methods

Results V: RfaH Family Analysis

Selecting the interdomain residues



- **30 putative metamorphic orthologs** from InterPro IPR010215: each has a NusG in the same organism, a predicted autoinhibited αCTD, and an isolated CTD predicted as β-barrel.
- **SRFI:** only 2 consistently frustrated residues, both outside the metamorphic region (110–162).
- **Contacts mode:** a group of highly conserved, minimally frustrated contacts at the NTD–αCTD interface, consistent with interdomain contacts stabilizing the α state.

Interface residue criteria

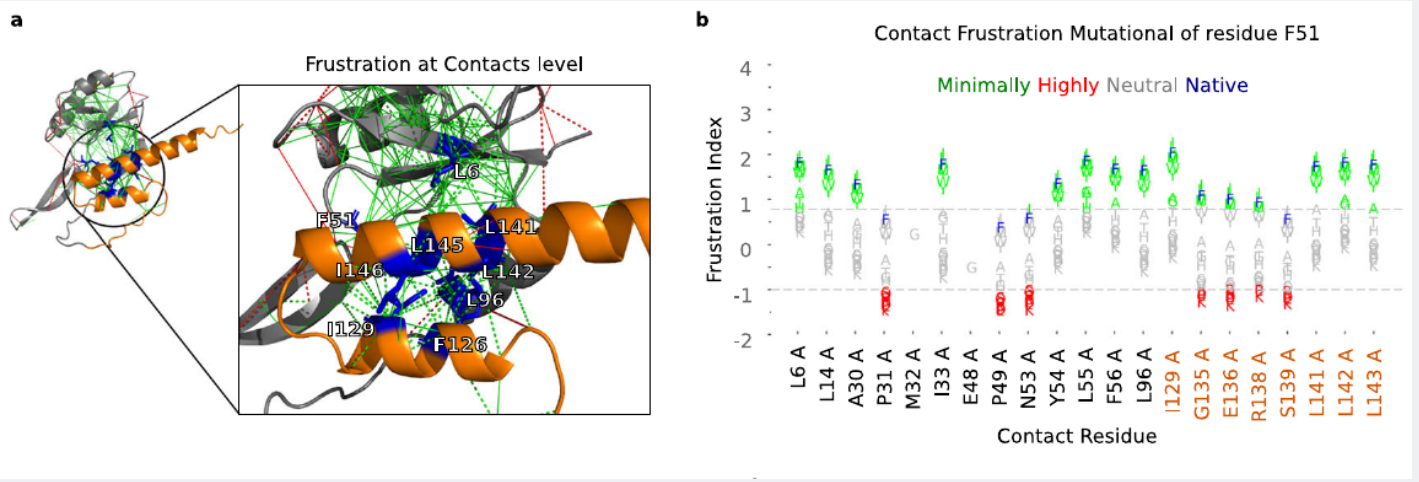
Interdomain contact; in the metamorphic region with contact FrustIC ≥ 0.5; in > 50% of models; > 50% of interactions minimally frustrated; ≥ 3 minimal contacts with the CTD.

→ **L6, F51, L96, F126, I129, L141, L142, L145, I146**

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Supplementary Fig. 12 and Methods

Results V: Mutational Scanning of the Interface

Similar- vs highly-frustrating substitutions



- F51 forms 21 contacts: 16 minimally frustrated, 5 neutral.
- F51M keeps them; F51K turns them neutral or highly frustrated.

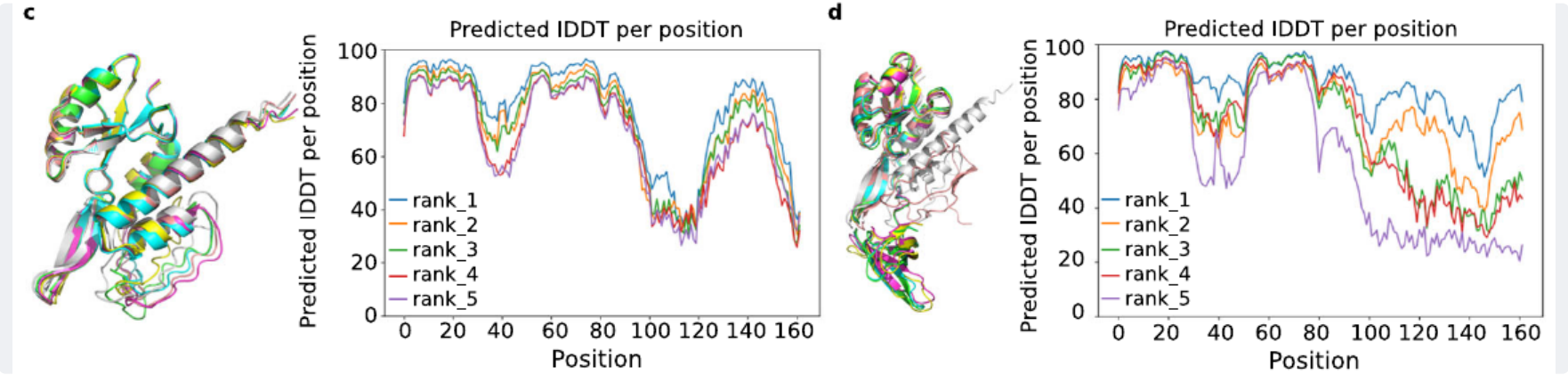
Set	Substitutions (one per interface residue)
SFMs	L6I, F51M, L96W, F126W, I129V, L141V, L142V, L145M, I146V
HFMs	L6D, F51K, L96K, F126N, I129E, L141D, L142K, L145E, I146D

SFM = similar frustration mutation (keeps the native stabilizing contacts); HFM = highly frustrated mutation (maximizes local frustration).

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Fig. 5a–b

Results V: Test 1, Designed Mutant Sets

Does frustrating the interface flip the predicted fold?



c · Keep frustration low

All 9 interface residues swapped for amino acids that keep their contacts minimally frustrated (e.g., F51M).

→ **AlphaFold2 still predicts the α-hairpin CTD, like normal RfaH.**

d · Add frustration

All 9 swapped for the amino acid that frustrates their contacts most (e.g., F51K).

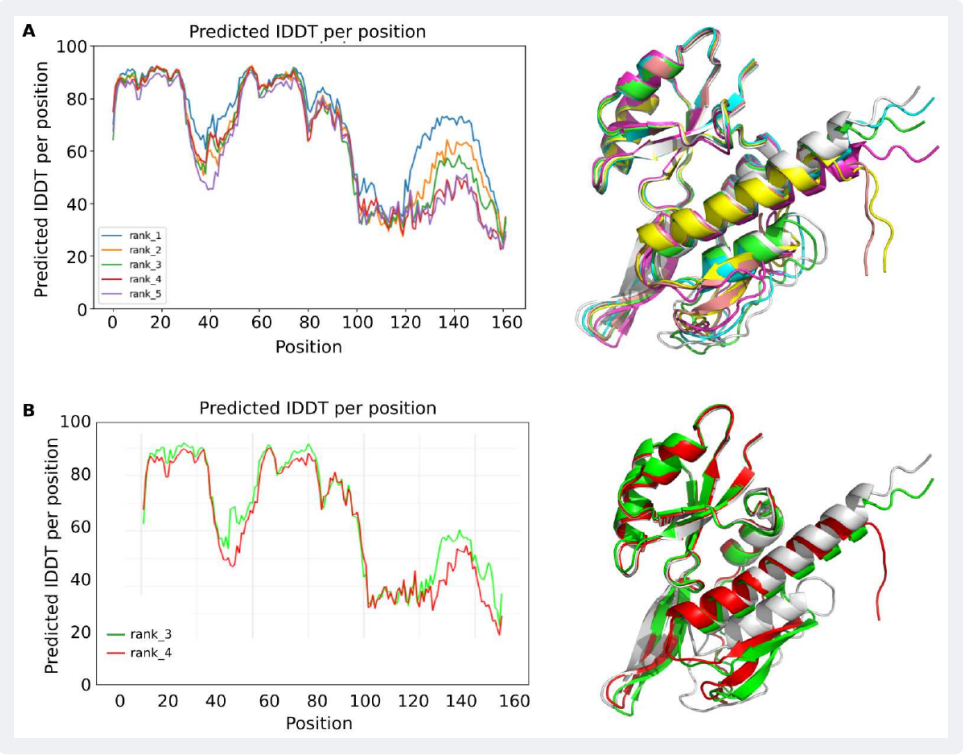
→ **AlphaFold2 predicts a β-barrel-like CTD, the active fold.**

Meaning: the conserved, minimally frustrated interface contacts are what hold the CTD in its α state. Breaking them pushes the prediction to the β fold (in silico only).

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Fig. 5c–d

Results V: Test 2, Learning from NusG

Which natural RfaH → NusG differences matter?



A: all 6 NusG-like mutations. B: L142S alone. Five AlphaFold2 models each.

- **Idea:** NusG, RfaH's relative, only has the β fold. How does it differ at the 9 interface residues?
- **6 of 9 differ:** L6V, I129V, L141V, L145I, I146F, L142S.
- **Frustration check:** 4 keep the contacts minimally frustrated, I146F makes them neutral, and only **L142S** frustrates strongly.
- **All 6 in RfaH (A):** 4 of 5 models show a β-like CTD.
- **L142S alone (B):** mixed, 1 α, 2 loop-like, 2 β. Of all 115 RfaH/NusG differences, it is the only single change that does this.

Meaning: adding frustration at Leu142 starts the switch, but also tuning the minimal contacts gives a more NusG-like CTD. Such patterns could guide mutations that lock or trigger a fold.

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Supplementary Figs. 14–15

Summary and Outlook

Impact, limitations and future directions

Methodological impact

- A physics-based reading of sequence conservation in an evolutionary context.
- Separates fold-level stability constraints from localized functional adaptations.

Current limitations

- **Static input:** one structure per protein; flexible or multi-state proteins need ensembles.
- **Thresholds:** three discrete classes can hide subtle continuous shifts.
- **MSA dependency:** curated alignments and subfamilies avoid blurred signals.

Future horizons

- Ensemble-aware inputs (e.g., MD or conformational-ensemble prediction).
- Database-scale annotation (CATH, PDBe-KB, AlphaFold DB).
- Family constraints as context for disease variant effects.

Source: Freiberger et al., *Nat. Commun.* 14, 8379 (2023), Discussion

Thank you for listening...

Code: github.com/proteinphysiologylab/FrustraEvo · Data: doi.org/10.5281/zenodo.10093060